Bài tập lần 14

Sách Pinchover - Rubinstein, Trang 128

5.15

$$(I)\begin{cases} u_{tt} - c^2 u_{xx} = F(x,t), & 0 < x < L,\ t > 0 \\ u_x(0,t) = t^2,\ u(L,t) = -t & ,\ t \geq 0 \\ u(x,0) = x^2 - L^2 \\ u_t(x,0) = \sin^2 \frac{\pi x}{L} & ,\ 0 \leq x \leq L \end{cases}$$

- Giả sử u_1 và u_2 là 2 nghiệm của (I).

Đặt $v = u_1 - u_2$. Ta sẽ CMR $v \equiv 0$

Ta có $v_{tt} = u_{1tt} - u_{2tt}$

$= (c^2 u_{1xx} + F) - (c^2 u_{2xx} + F)$

$= c^2(u_{1xx} - u_{2xx}) = c^2 v_{xx}$ $\quad 0 < x < L,\ t > 0$

$v_x(x,t) = u_{1x}(x,t) - u_{2x}(x,t)$

$\Rightarrow v_x(0,t) = u_{1x}(0,t) - u_{2x}(0,t) = t^2 - t^2 = 0$

$v(L,t) = u_1(L,t) - u_2(L,t) = -t + t = 0$

$v(x,0) = u_1(x,0) - u_2(x,0) = (x^2 - L^2) - (x^2 - L^2) = 0$

$v_t(x,0) = u_{1t}(x,0) - u_{2t}(x,0) = \sin^2\frac{\pi x}{L} - \sin^2\frac{\pi x}{L} = 0$

$$\rightarrow \begin{cases} v_{tt} = c^2 v_{xx} & 0 < x < L,\ t > 0 \\ v_x(0,t) = v(L,t) = 0 & t \geq 0 \\ v(x,0) = v_t(x,0) = 0 & 0 \leq x \leq L \end{cases}$$

Xét $I(t) = \frac{1}{2}\int_0^L [v_t^2(x,t) + c^2 v_x^2(x,t)]\,dx$.

$I(0) = \frac{1}{2}\int_0^L [v_t^2(x,0) + a^2 v_x^2(x,0)]\,dx = \frac{1}{2}\int_0^L [0^2 + a^2 0^2]\,dx = 0$

$I'(t) = \int_0^L (v_t v_{tt} + c^2 v_x v_{xt})\,dx$

$= \int_0^L (c^2 v_t v_{xx} + c^2 v_x v_{xt})\,dx = c^2\int_0^L (v_t v_x)'_x\,dx$

$$= c^2 v_t v_x \Big|_{x=0}^{x=L}$$

$$= c^2 \left[v_t(L,t)\, v_x(L,t) - v_t(0,t)\, v_x(0,t) \right] = 0$$

(Do $v(L,t) = v_x(0,t) = 0 \Rightarrow v_t(L,t) = v_x(0,t) = 0$.)

$$\to I(t) = I(0) = \int_0^L \left(v_t^2 + c^2 v_{xx} \right)$$

$$= \frac{1}{2} \int_0^L \left(v_t^2 + c^2 v_x^2 \right) dx = 0$$

$$\to v_t^2 + c^2 v_x^2 = 0 \quad \forall x \in [0, L]$$

$$\to v_t = v_x = 0 \to v = \text{Const} = v(x,0) = 0$$

$$\to u_1 = u_2$$

→ Bài toán (I) có nghiệm duy nhất u (Nếu u là nghiệm của bài toán)

5.16

$$(I) \begin{cases} u_{tt} + u_t - c^2 u_{xx} = 0 & a < x < b,\ t > 0 \\ u(a,t) = u_x(b,t) = 0 & t \geq 0 \\ u(x,0) = f(x) & a \leq x \leq b \\ u_t(x,0) = g(x) & a \leq x \leq b \end{cases}$$

- Giả sử u_1 và u_2 là 2 nghiệm của (I)

Đặt $v = u_1 - u_2$. Ta sẽ CMR $v = 0$

Ta có $v_{tt} = u_{1tt} - u_{2tt}$

$$= (c^2 u_{1xx} - u_{1t}) - (c^2 u_{2xx} - u_{2t})$$

$$= c^2 (u_{1xx} - u_{2xx}) - (u_{1t} - u_{2t})$$

$$= c^2 v_{xx} - v_t$$

$$\to v_{tt} + v_t = c^2 v_{xx} \quad (a < x < b,\ t > 0)$$

$$v(a,t) = u_1(a,t) - u_2(a,t) = 0$$

$$v_x(b,t) = u_{1x}(b,t) - u_{2x}(b,t) = 0$$

$$v(x,0) = u_1(x,0) - u_2(x,0) = f(x) - f(x) = 0$$

$v_t(x,0) = u_{1t}(x,0) - u_{2t}(x,0) = g(x) - g(x) = 0$

$$\begin{cases} v_{tt} + v_t = c^2 v_{xx} & a < x < b,\ t > 0 \\ v(a,t) = v_x(b,t) = 0 & t \geq 0 \\ v(x,0) = v_t(x,0) = 0 & a \leq x \leq b \end{cases}$$

Xét $I(t) = \frac{1}{2}\int_a^b (v_t^2 + c^2 v_x^2)\,dx$

$I(0) = \frac{1}{2}\int_a^b (v_t^2(x,0) + c^2 v_x^2(x,0))\,dx = 0$

(Do $v(x,0) = v_t(x,0) = 0 \Rightarrow v_x(x,0) = v_t(x,0) = 0$)

$I'(t) = \int_a^b (v_t v_{tt} + c^2 v_x v_{xt})\,dx$

$= \int_a^b v_t(c^2 v_{xx} - v_t + c^2 v_x v_{xt})\,dx = \int_a^b [c^2(v_t v_{xx} + v_x v_{xt}) - v_t^2]\,dx$

$= c^2\int_a^b (v_t v_{xx} + v_x v_{xt})\,dx - \int_a^b (v_t)^2\,dx$

$= c^2(v_t v_x)\Big|_{x=a}^{x=b} - \int_a^b (v_t)^2\,dx$

$= c^2 v_t(b,t)v_x(b,t) - c^2 v_t(a,t)v_x(a,t) - \int_a^b (v_t)^2\,dx = 0 - 0 - \int_a^b (v_t)^2\,dx$

(Do $v(a,t) = v_x(b,t) = 0 \Rightarrow v_t(a,t) = v_x(b,t) = 0$)

$= -\int_a^b (v_t)^2\,dx \leq 0$

$\rightarrow\ 0 \leq I(t) \leq I(0) = 0 \Rightarrow I(t) = 0 = \frac{1}{2}\int_a^b (v_t^2 + c^2 v_x^2)\,dx\ \forall x \in [a,b]$

$\rightarrow v_t = v_x = 0 \Rightarrow v = \text{const} = v(x,0) = 0$

$\rightarrow u_1 = u_2$

$\rightarrow$ Nếu u là n° của (I) thì u là duy nhất.

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Ngày

5.17

$$(I)\begin{cases} u_{tt} - c^2u_{xx} + hu = F(x,t) & -\infty < x < +\infty,\ t>0,\ h>0 \\ \lim\limits_{x\to\pm\infty} u(x,t) = \lim\limits_{x\to\pm\infty} u_x(x,t) = \lim\limits_{x\to\pm\infty} u_t(x,t) = 0 & t\geq 0 \\ \int_{-\infty}^{+\infty} (u_t^2 + c^2u_x^2 + hu^2)\,dx < \infty & ,t\geq 0 \\ u(x,0) = f(x) & -\infty < x < +\infty \\ u_t(x,0) = g(x) \end{cases}$$

- Giả sử u_1 và u_2 là nghiệm của (I), Ta CM
Đặt $v = u_1 - u_2$. Ta CMR $v \equiv 0$

Ta có $v_{tt} = u_{1tt} - u_{2tt}$

$$= c^2u_{1xx} - hu_1 + F - c^2u_{2xx} + hu_2 - F$$

$$= c^2(u_{1xx} - u_{2xx}) - (hu_1 - hu_2)$$

$$= c^2v_{xx} - hv$$

$$\Rightarrow v_{tt} + hv - c^2v_{xx} = 0$$

$$\lim_{x\to\pm\infty} v(x,t) = \lim_{x\to\pm\infty} (u_1(x,t) - u_2(x,t)) = 0$$

$$\lim_{x\to\pm\infty} v_x(x,t) = \lim_{x\to\pm\infty} (u_{1x}(x,t) - u_{2x}(x,t)) = 0$$

$$\lim_{x\to\pm\infty} v_t(x,t) = \lim_{x\to\pm\infty} (u_{1t}(x,t) - u_{2t}(x,t)) = 0$$

$$v(x,0) = u_1(x,0) - u_2(x,0) = f(x) - f(x) = 0$$

$$v_t(x,0) = u_{1t}(x,0) - u_{2t}(x,0) = g(x) - g(x) = 0$$

$$\Rightarrow \begin{cases} v_{tt} - c^2v_{xx} + hv = 0 & -\infty < x < +\infty,\ t>0 \\ \lim\limits_{x\to\pm\infty} v(x,t) = \lim\limits_{x\to\pm\infty} v_x(x,t) = \lim\limits_{x\to\pm\infty} v_t(x,t) = 0 & t\geq 0 \\ v(x,0) = v_t(x,0) = 0 & -\infty < x < +\infty \end{cases}$$

Thứ
Ngày . . No.

Xét $I(t) = \frac{1}{2}\int_{-\infty}^{+\infty} (v_t^2 + c^2 v_x^2 + h v^2)\, dx$

$I(0) = \frac{1}{2}\int_{-\infty}^{+\infty} (\underset{0}{\underset{\shortparallel}{v_t^2(x,0)}} + c^2 \underset{0}{\underset{\shortparallel}{v_x^2(x,0)}} + h \underset{0}{\underset{\shortparallel}{v^2(x,0)}})\, dx = 0$

$I'(t) = \int_{-\infty}^{+\infty} (v_t v_{tt} + c^2 v_x v_{xt} + h v v_t)\, dx.$

$= \int_{-\infty}^{+\infty} (v_t(-hv + c^2 v_{xx}) + c^2 v_x v_{xt} + h v v_t)\, dx$

$= c^2 \int_{-\infty}^{+\infty} (v_t v_{xx} + v_x v_{xt})\, dx = c^2 \int_{-\infty}^{+\infty} (v_t v_x)'_x\, dx.$

$= \lim_{\substack{\xi \to +\infty \\ \mu \to -\infty}} c^2 (v_t v_x)\Big|_{x=\mu}^{x=\xi}$

$= \lim_{\substack{\xi \to +\infty \\ \mu \to -\infty}} c^2 [v_t(\xi,t) v_x(\xi,t) - v_t(\mu,t) v_x(\mu,t)]$

$= 0$ Do $\lim_{x \to \pm\infty} v_x(x,t) = 0$

$\to I'(t) = \text{const} = I(0) = 0$

$\to I(t) = \text{const} = I(0) = 0$

$\to \frac{1}{2}\int_{-\infty}^{+\infty} (v_t^2 + c^2 v_x^2 + h v^2)\, dx = 0$

$\to v_t^2 + c^2 v_x^2 + h v^2 = 0$

$\to v_t = v_x = v = 0$

$\to u_1 = u_2$

$\Rightarrow$ Kết luận: Nếu u là 1 n° của (I) thì u là duy nhất

5.18

$$(I)\begin{cases} u_t - k u_{xx} = F(x,t), & 0 < x < L,\ t > 0 \\ u(0,t) - \alpha u_x(0,t) = a(t) & t \geq 0 \\ u(L,t) + \beta u_x(L,t) = b(t) \\ u(x,0) = f(x) - & 0 < x < L \end{cases}$$

- Giả sử u_1 và u_2 là 2 n° của (I)

Đặt $v = u_1 - u_2$. Ta CMR $v \equiv 0$

Ta có $v_t = u_{1t} - u_{2t}$

$$= k u_{1xx} + F(x,t) - k u_{2xx} - F(x,t)$$

$$= k(u_{1xx} - u_{2xx}) = k v_{xx}$$

$$v(0,t) = u_1(0,t) - u_2(0,t)$$

$$= \alpha u_{1x}(0,t) + a(t) - \alpha u_{2x}(0,t) - a(t)$$

$$= \alpha(u_{1x}(0,t) - u_{2x}(0,t)) = \alpha v_x(0,t)$$

$$v(L,t) = u_1(L,t) - u_2(L,t)$$

$$= b(t) - \beta u_{1x}(L,t) - b(t) + \beta u_{2x}(L,t)$$

$$= -\beta v_x(L,t)$$

$$v(x,0) = u_1(x,0) - u_2(x,0) = f - f = 0$$

$$\rightarrow \begin{cases} v_t = k v_{xx} & 0 < x < L,\ t > 0 \\ v(0,t) = \alpha v_x(0,t),\ v(L,t) = -\beta v_x(L,t) & t > 0 \\ v(x,0) = 0 & 0 < x < L \end{cases}$$

Xét $I(t) = \frac{1}{2}\int_0^L v^2(x,t)\,dx$

$$I(0) = \frac{1}{2}\int_0^L v^2(x,0)\,dx = 0$$

$$I'(t) = \int_0^L v v_t\,dx = k\int_0^L v\, v_{xx}\,dx$$

$$= k\int_0^L v\, d(v_x) = k\left(v v_x \Big|_{x=0}^{x=L} - \int_0^L (v_x)^2 dx \right)$$

$$= k\left(v(L,t)v_x(L,t) - v(0,t)v_x(0,t) - \int_0^L v_x^2(x,t)\,dx \right)$$

$$= k\left(-\beta \underbrace{v_x^2(L,t)}_{\geq 0} - \alpha \underbrace{v_x^2(0,t)}_{\geq 0} - \int_0^L \underbrace{v_x^2(x,t)}_{\geq 0}\,dx \right) \quad (\alpha, \beta \geq 0),\ k > 0$$

≤ 0

$\Rightarrow 0 \leq I(t) \leq I(0) = 0$

$\rightarrow I(t) = 0 = \frac{1}{2}\int_0^L v^2(x,t)\,dx$

$\rightarrow v(x,t) = 0 \rightarrow u_1 = u_2$

$\Rightarrow$ Nếu u là nº của (I) thì u là duy nhất.

Câu 3 ý (a) Đề 3 – GK – K64 TT

$$(I)\begin{cases} u_t(x,t) = 3u_{xx}(x,t) - 6u_x(x,t) + e^x \sin x & 0 < x < \pi, t > 0 \\ u(0,t) = u(\pi,t) = 0 & t \geq 0 \\ u(x,0) = xe^x. & 0 \leq x \leq \pi \end{cases}$$

– Giả sử u_1 và u_2 là 2 nº của (I)

Đặt $v = u_1 - u_2$. Ta CMR $v \equiv 0$

Ta có $v_t = u_{1t} - u_{2t} =$

$= 3u_{1xx} - 6u_{1x} + e^x \sin x - 3u_{2xx} + 6u_{2x} - e^x \sin x$

$= 3(u_{1xx} - u_{2xx}) - 6(u_{1x} - u_{2x})$

$= 3v_{xx} - 6v_x$

$\rightarrow v_t = 3v_{xx} - 6v_x$

$v(0,t) = u_1(0,t) - u_2(0,t) = 0$

$v(\pi,t) = u_1(\pi,t) - u_2(\pi,t) = 0$

$v(x,0) = u_1(x,0) - u_2(x,0) = xe^x - xe^x = 0$

$$\rightarrow \begin{cases} v_t = 3v_{xx} - 6v_x & 0 < x < \pi, t > 0 \\ v(0,t) = v(\pi, t) = 0 & t \geq 0 \\ v(x,0) = 0 & 0 \leq x \leq \pi \end{cases}$$

Xét $I(t) = \frac{1}{2}\int_0^{\pi} v^2(x,t)\,dx$

$\rightarrow I(0) = \frac{1}{2}\int_0^{\pi} v^2(x,0)\,dx = 0$

$I'(t) = \int_0^{\pi} v v_t\,dx = \int_0^{\pi} v(3v_{xx} - 6v_x)\,dx$

$= 3\int_0^{\pi} v v_{xx}\,dx - 6\int_0^{\pi} v v_x\,dx$

Xét $I_1(t) = 3\int_0^{\pi} v(x,t)\,v_{xx}(x,t)\,dx$

$= 3\left(v(x,t)\,v_x(x,t)\Big|_0^{\pi} - \int_0^{\pi} v_x^2(x,t)\,dx\right)$

$= 3\left(v(\pi,t)\,v_x(\pi,t) - v(0,t)\,v_x(0,t) - \int_0^{\pi} v_x^2(x,t)\,dx\right)$

$= -3\int_0^{\pi} v_x^2(x,t)\,dx$

Xét $I_2(t) = 6\int_0^{\pi} v(x,t)\,v_x(x,t)\,dx$

$= 6\left(v^2(x,t)\Big|_0^{\pi} - \int_0^{\pi} v(x,t)\,v_x(x,t)\,dx\right)$

$= 6\left(v^2(\pi,t) - v^2(0,t)\right) - 6I_2(t)$

$\rightarrow 7I_2(t) = 0 \rightarrow I_2(t) = 0$

$\Rightarrow I'(t) = I_1(t) - I_2(t)$

$$= -3\int_0^{\pi} v_x^2(x,t)\,dx \le 0$$

$\Rightarrow 0 \le I(t) \le I(0) = 0$

$\Rightarrow I(t) = 0 = \frac{1}{2}\int_0^{\pi} v^2(x,t)\,dx$

$\Rightarrow v(x,t) = 0 \Rightarrow u_1 = u_2$

$\Rightarrow$ Vậy bài toán (I) có tối đa 1 nghiệm

Câu 3 - ý (a) đề 5 - GK - K64 π

$$(I)\begin{cases} u_{tt} = u_{xx} - u + F(x,t) & 0 < x < 2, t > 0 \\ u_x(0,t) = 2, \; u(2,t) = e^2+1 & t \ge 0 \\ u(x,0) = e^x + 1, \; u_t(x,0) = \cos^3\frac{\pi x}{4} & 0 \le x \le 2 \end{cases}$$

- Giả sử u_1 và u_2 là 2 n° của (I)

Đặt $v = u_1 - u_2$. Ta CMR $v \equiv 0$

Ta có $v_{tt} = u_{1tt} - u_{2tt}$

$= u_{1xx} - u_1 + F(x,t) - u_{2xx} + u_2 - F(x,t)$

$= (u_{1xx} - u_{2xx}) - (u_1 - u_2) = v_{xx} - v.$

$v_x(0,t) = u_{1x}(0,t) - u_{2x}(0,t) = 2 - 2 = 0$

$v(2,t) = u_1(2,t) - u_2(2,t) = e^2+1 - e^2 - 1 = 0$

$v(x,0) = u_1(x,0) - u_2(x,0) = e^x + 1 - e^x - 1 = 0$

$v_t(x,0) = u_{1t}(x,0) - u_{2t}(x,0) = \cos^3\frac{\pi x}{4} - \cos^3\frac{\pi x}{4} = 0$

$$\Rightarrow \begin{cases} v_{tt} = v_{xx} - v & 0 < x < 2, t > 0 \\ v_x(0,t) = v(2,t) = 0 & t \ge 0 \\ v(x,0) = v_t(x,0) = 0 & 0 \le x \le 2 \end{cases}$$

Xét $I(t) = \frac{1}{2}\int_0^2 (v_t^2 + v_x^2 + v^2)\,dx.$

Thứ

Ngày

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$\to I(0) = \frac{1}{2}\int_0^2 \left(v_t^2(x,0) + v_x^2(x,0) + v^2(x,0)\right) dx = 0$

$I'(t) = \int_0^2 (v_t v_{tt} + v_x v_{xt} + v v_t)\, dx$

$= \int_0^2 (v_t(v_{xx} - v) + v_x v_{xt} + v v_t)\, dx$

$= \int_0^2 (v_t v_{xx} + v_x v_{xt})\, dx = \int_0^2 (v_t v_x)'_x\, dx$

$= v_t v_x \Big|_{x=0}^{x=2} = v_t(2,t) v_x(2,t) - v_t(0,t) v_x(0,t) = 0$

Do $v_x(0,t) = v(2,t) = 0 \Rightarrow v_x(0,t) = v_t(2,t) = 0$

$\Rightarrow I(t) = I(0) = 0 = \frac{1}{2}\int_0^2 (v_t^2 + v_x^2 + v^2)\, dx$

$\to v_t = v_x = v = 0$

$\Rightarrow u_1 = u_2$

$\Rightarrow$ Bài toán (I) có tối đa 1 n°

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